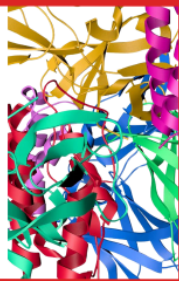


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Critical Role of TLR9 in Acute Graft-versus-Host Disease¹

Claudia Calcaterra,^{2*} Lucia Sfondrini,^{2†} Anna Rossini,[†] Michele Sommariva,[‡]
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Graft-vs-host disease (GVHD) is a major complication after allogeneic bone marrow transplantation. Different studies have demonstrated that intestinal bacterial breakdown products and loss of gastrointestinal tract integrity, both induced by conditioning regimens, are critical in the pathogenesis of acute GVHD. Using C57BL/6 knockout mice, we evaluated the role of TLR4 and TLR9, which recognize bacterial LPS and DNA, respectively, in the GVHD associated with allogeneic bone marrow transplantation. When myeloablative-irradiated TLR9 knockout (TLR9^{-/-}) mice were used as graft recipients, survival and clinical score of acute GVHD were improved as compared with the wild-type recipient mice (18/30 vs 1/31 mice still alive at day 70 in a total of four experiments); while no differences were observed using recipient TLR4 knockout (TLR4^{-/-}) mice. The reduced mortality and morbidity in TLR9^{-/-} mice related with reduced stimulatory activity of TLR9^{-/-} spleen APCs after conditioning and reduced proliferation of allogeneic donor T cells. Experiments using TLR9^{+/+} into TLR9^{-/-} and TLR9^{-/-} into TLR9^{+/+} chimeric mice as recipients indicated a critical role for nonhematopoietic TLR9^{+/+} cells interacting with bacterial breakdown products released in myeloablated mice. Altogether these data reveal a novel important role of TLR9 in GVHD, a finding that might provide tools to reduce this complication of allogeneic transplantation. *The Journal of Immunology*, 2008, 181: 6132–6139.

Allogeneic bone marrow transplantation (BMT)⁴ following myeloablative conditioning therapy is a useful treatment for a variety of hematologic malignancies and severe immunodeficiencies. However, its use is restricted by the serious effects of acute graft-vs-host disease (GVHD) caused by activated donor T cells that recognize major and minor histocompatibility Ag mismatches and thereby induce an inflammatory process targeting recipient tissues. A variety of studies have suggested a role for intestinal bacterial microflora in the pathogenesis of acute GVHD, including the demonstration in germ-free or completely decontaminated rodents that the absence or complete growth suppression of intestinal bacteria prevents the development of acute GVHD in recipient animals of MHC-mismatched transplants (1, 2), the reported amelioration of GVHD following modification of intestinal flora by oral administration of probiotics (3), and the finding that antimicrobial chemotherapy targeted to intestinal anaerobic bacteria in human marrow transplant recipients significantly reduces the severity of acute GVHD (4). Hence, bowel

decontamination using broad-spectrum antibiotics before transplantation has been introduced as standard practice to date (4, 5).

Conserved molecular patterns (pathogen-associated molecular patterns) of microorganisms are recognized by a variety of pattern-recognition receptors that include the TLRs family. Depending on the type of cell, TLR recognition of microbial products elicits a cascade of signaling transduction pathways, resulting in the production of proinflammatory chemokines and cytokines, antimicrobial peptides, adhesion molecules, and costimulatory molecules that regulate the activation of both innate and adaptive immunity. Different immune cell types, as dendritic cells, macrophages, monocytes, B cells, mast cells, NK cells, regulatory T cells, and neutrophils have been reported to express TLRs (6, 7). Recent studies have described expression of TLRs also in nonprofessional immune cells, as airway and intestinal epithelial cells and endothelial cells (6, 8, 9).

Few and controversial studies have not well-defined yet the role of TLR activation on GVHD (10–12). Since the TLR4 ligand LPS, one of microbial breakdown products produced by conditioning regimens, has been demonstrated in serum after BMT with prior myeloablative conditioning therapy (13, 14), in human studies the relationship between TLR signaling and GVHD focused on TLR4. One study analyzing data from retrospective studies in humans indicated a reduced risk of acute GVHD associated with TLR4 mutations (15), whereas another study reported an increased risk (16). However, since microbial breakdown products produced by conditioning regimens could also include the TLR9 ligand CpG-containing DNA and TLR9, which is expressed on immune and intestinal epithelial cells (17–19), has been reported to play a unique role in maintenance of colonic homeostasis and inflammation and in regulating tolerance to other TLR ligands (20, 21), in this study, we examined the role of TLR4 and TLR9 in a murine model of GVHD associated with allogeneic BMT.

Materials and Methods

Mice

BALB/c (H2d) and C57BL/6 (B6, H2b) mice were purchased from Charles River Laboratories. C57BL/6 mice knocked out for TLR9 (TLR9^{-/-}) or

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⁴ Abbreviations used in this paper: BMT, bone marrow transplantation; GVHD, graft-vs-host disease; ODN, oligonucleotide.

TLR4 (TLR4^{-/-}) expression, respectively, were both originally obtained from Shizuo Akira (Osaka University, Japan) (22, 23). C57BL/6 mice knocked out for both TLR9 and TLR4 (TLR9/4^{-/-}) were obtained by mating TLR9^{-/-} and TLR4^{-/-} mice and housed in our animal facility. All mice were female and 8–12 wk old (20–24 g body weight) at the start of the experiments. Mice were maintained at constant temperature and humidity, with food and water given ad libitum. Experimental protocols were approved by the Ethics Committee for Animal Experimentation of the Istituto Nazionale Tumori of Milan.

Induction of GVHD

C57BL/6 wild-type, TLR9^{-/-} or TLR4^{-/-} or TLR9/4^{-/-} mice received myeloablative total-body irradiation (900 cGy, ¹³⁷Cs source) in two doses with a 2-h interval to minimize gastrointestinal toxicity, followed by i.v. infusion of 1×10^7 donor BALB/c bone marrow cells and 4×10^7 or 5×10^7 BALB/c splenocytes as a source of allogeneic T cells. The reverse experiments were performed using irradiated BALB/c mice as recipients and C57BL/6 or TLR9^{-/-} or TLR4^{-/-} mice as donors of 1×10^7 bone marrow cells and 4×10^7 splenocytes. Bone marrow cell suspensions were prepared by flushing femurs and tibiae with RPMI 1640 medium supplemented with 10% FCS, nonessential amino acids, L-glutamine, sodium pyruvate, and 2-ME; splenocyte preparations were obtained by gently crushing the spleen in complete medium to release the cells. RBC were lysed by incubation in ammonium chloride lysing solution (0.15 M NH₄Cl, 10 mM KHCO₃, and 1 mM Na₄EDTA (pH 7.2)) for 10 min at 4°C and preparations were filtered to remove debris and washed twice in PBS for injection.

In experiments including antibiotic treatment to prevent and treat infections induced by conditioning regimens, mice received an oral suspension of gentamicin (70 mg/l) in their drinking water beginning 7 days before transplantation and continuing until 2 mo after BMT. Gentamicin is a non-absorbable broad-spectrum antibacterial agent, reported to induce reduction of strict anaerobic bacteria and lactose fermenting organisms, resulting strongly effective against bacteremia from endogenous organisms, one of the major reasons of high mortality after total body irradiation (24, 25).

In all experiments, survival was monitored daily and recipients' body weights and GVHD clinical score were assessed twice weekly.

To generate TLR9^{+/+} into TLR9^{-/-} and TLR9^{-/-} into TLR9^{+/+} chimeras, TLR9^{-/-} and C57BL/6 wild-type (TLR9^{+/+}) mice were lethally irradiated and reconstituted with 1×10^7 bone marrow cells and 4×10^7 splenocytes obtained by TLR9^{+/+} or TLR9^{-/-} mice, respectively. A control group was obtained by reconstituting irradiated wild-type mice with autologous bone marrow cells and splenocytes. To verify engraftment, PB-MCs were analyzed at 6 wk posttransplantation for the expression of TLR9 by PCR using specific primers: 5'-TAT AGG ACA CCA GGA GGT ACT C-3' and 5'-AAC ATG GTT CTC CGT CGA AGG A-3'. A 40-cycle PCR was run in a GeneAmp 9700 (Applied Biosystems) using the following profile: 94°C for 60 s, 60°C for 60 s, and 72°C for 60 s. Chimeric and control mice were then subjected to myeloablative irradiation, followed by i.v. infusion of 1×10^7 donor allogeneic BALB/c bone marrow cells and 4×10^7 BALB/c splenocytes.

Assessment of GVHD

GVHD severity was assessed using the previously described clinical scoring system (26). Each BMT recipient mouse was scored twice weekly for five parameters: weight loss, posture (hunching), activity, fur texture, and skin integrity using a scale of 0 to 2, with 0 for absent or normal, 1 for mildly abnormal, and 2 for severely abnormal. The GVHD clinical index was the sum of the scores for individual criteria (maximum index = 10) (26).

Histology

Samples of small and large intestine, liver, and skin (from the abdomen) from C57BL/6 wild-type and TLR9^{-/-} mice (two mice/group) were collected 7 days after allogeneic transplantation with 1×10^7 donor BALB/c bone marrow cells and 4×10^7 BALB/c splenocytes. Samples were fixed in 10% formalin, embedded in paraffin, cut into 5 μ m-thick sections, and stained with H&E. Tissue sections were examined for evidence of GVHD as described (27). Images were obtained with a digital camera mounted under a light microscope.

CFSE labeling

Ex vivo experiments. To evaluate the proliferation activity of allogeneic donor BALB/c T cells in TLR9^{-/-} and C57BL/6 transplanted mice, spleen cells were aseptically removed at day 4 after BMT and single-cell suspensions of T cell-enriched splenocytes were prepared by negative selection

with mouse CD19 microbeads (Miltenyi Biotec) to remove proliferating B cells and to enrich T cells. Briefly, RBC were lysed and single-cell suspensions of splenocytes were magnetically labeled with the mouse CD19 microbeads for 15 min at 4°C, washed, and passed through a separation MS+ column placed in the magnetic field of a MACS separator (Miltenyi Biotec). Unlabeled cells passing through the column were collected, labeled in 5 μ M CFSE according to the manufacturer's recommendations (Molecular Probes), and cultured in complete RPMI 1640 at a final concentration of 1×10^6 cells/ml in 96-well plates. To determine appropriate in vitro conditions to evaluate proliferation after CFSE labeling, dose- and time-defining experiments were performed. Just before analysis, cells were stained with PE-conjugated anti-H2D^d Ab (BD Pharmingen) for 25 min on ice. Proliferation of donor H2d cells was evaluated after 3 days of culture, when two clearly distinct fractions of proliferating (CFSE-"low") and non-proliferating (CFSE-"high") were detectable. Flow cytometric analysis was performed using FACSCalibur and CellQuest software (BD Biosciences).

In vivo experiments. Proliferation of alloreactive donor T cells was evaluated in TLR9^{-/-} and C57BL/6 wild-type mice, lethally irradiated, and reconstituted with 1×10^7 bone marrow cells and 4×10^7 splenocytes obtained from BALB/c, by injecting 1.5×10^7 CFSE-labeled BALB/c T cell enriched splenocytes at day 4 after BMT. After 24 h, spleen cells were harvested, stained with PE-conjugated anti-H2D^d Ab (BD Pharmingen), and analyzed by FACScan.

Flow cytometric analyses

The percentage of T cells in spleen cell sample obtained from transplanted mice was analyzed by staining with FITC-anti-mouse CD3 (BD Pharmingen). Chimerism was assessed on PBMC and splenocytes of transplanted mice by staining with FITC-anti-mouse H-2D^d or H-2D^b (BD Pharmingen) Abs. Expression of costimulatory molecules on APCs derived from C57BL/6 and TLR9^{-/-} mice 3 or 24 h after myeloablative irradiation were evaluated in cells stained with FITC-anti-mouse IAb, PE-anti-mouse CD80, and PE-anti-mouse CD86 (BD Pharmingen). In all experiments, cells were stained at 4°C for 25 min, washed, and analyzed using FACSCalibur and CellQuest software (BD Biosciences). APCs were purified from splenocytes by positive selection with mouse MHC class II microbeads (Miltenyi Biotec).

MLR cultures

Responder T cells (1×10^5 cells/well) were cocultured with MHC class II-positive stimulator cells (stimulator:responder ratio, 2:1, 1:1, or 1:2) in U-bottom microwell plates (Costar) at 37°C. Responder cells were prepared from BALB/c mouse splenocytes by negative selection with mouse CD19 microbeads (Miltenyi Biotec) to remove proliferating B cells and to enrich T cells. Stimulator cells were obtained by positive selection with mouse MHC class II microbeads (Miltenyi Biotec) from spleen cells of C57BL/6 or TLR9^{-/-} mice nonirradiated or lethally irradiated 24 h before. In experiments using nonirradiated mice, stimulator cells were irradiated (3000 cGy) before coculture. In experiments performed to evaluate the capability of TLR9^{-/-} APC to respond to maturation stimuli, LPS (1 μ g/ml) and/or TNF- α (10 ng/ml) were added to stimulator cells for 24 h before coculture. In all experiments, proliferation of responder cells was evaluated in the presence of stimulator cells obtained from three different mice per group. [³H]TdR (1 μ Ci) was added to each well (six wells/mouse) for the final 18 h of culture and [³H]TdR incorporation was evaluated in a scintillation counter. Proliferation of T cells was evaluated at different times of coculture with MHC class II stimulator cells (1–5 days).

Statistical analysis

Differences in survival curves were analyzed using the log-rank test. Differences in mean values of GVHD index score, in vitro cell proliferation, percentage of CFSE^{high}- or CFSE^{low}-H2D^d cells, and expression of costimulatory molecules in the different experimental groups were compared using two-tailed unpaired Student's *t* test. Differences were considered significant at *p* $\leq$ 0.05.

Results

Reduced acute GVHD severity in TLR9^{-/-} mice

Allogeneic BMT experiments were performed in C57BL/6 wild-type, TLR4^{-/-} and TLR9^{-/-} mice to evaluate the involvement of the two TLRs in the pathogenesis of acute GVHD. Mice (eight mice/group/strain) were myeloablative-irradiated and injected with 10^7 bone marrow cells and 4×10^7 splenocytes obtained from full MHC major and minor Ag-disparate BALB/c donors. Recipient

mice were monitored for clinical signs of GVHD, weight, and survival. All wild-type and TLR4^{-/-} mice succumbed to severe acute GVHD within 60 days, while TLR9^{-/-} mice showed a significantly higher survival rate, with four of eight mice still alive at the end of the experiment (day 70) (Fig. 1A; $p = 0.0062$ vs C57BL/6; $p = 0.0035$ vs TLR4^{-/-}). The GVHD clinical score in TLR9^{-/-} mice was also significantly lower than that in TLR4^{-/-} and C57BL/6 mice ($p < 0.0001$) (Fig. 1B), while no significant differences were observed considering weight loss alone (Fig. 1C). At the end of the experiment (day 70), all TLR9^{-/-} surviving mice showed 100% donor cells (data not shown). Absence of a significant involvement of TLR4 in the pathogenesis of GVHD was supported by results of allogeneic BMT experiments using TLR9/4 double-knockout mice (TLR9/4^{-/-}), whose survival rate (four of seven mice still alive) at day 70, when all wild-type mice were dead, was not significantly increased over that of TLR9^{-/-} mice (three of seven alive).

In subsequent allogeneic BMT experiments, in which gentamicin was added to the drinking water in an effort to reduce mortality related to posttransplantation infections, 11 of 15 TLR9^{-/-} mice were still alive at day 70 vs 1 of 15 wild-type mice (Fig. 2A; $p < 0.0001$ TLR9^{-/-} vs C57BL/6). Again, the GVHD clinical score in TLR9^{-/-} mice was significantly lower than that in wild-type mice ($p < 0.0001$) (Fig. 2B), while no significant difference was observed considering weight loss alone (Fig. 2C). When the number of injected allogeneic splenocytes was increased to 5×10^7 , all five wild-type mice succumbed to GVHD within 31 days (mean survival time: 17.2 days), while two of five TLR9^{-/-} mice were still alive at day 70 (mean survival time 59.8 days) ($p = 0.0023$).

The reduced clinical signs of GVHD in TLR9^{-/-} mice correlated with reduced intestinal damage. Indeed, small intestine samples collected from C57BL/6 wild-type mice or TLR9^{-/-} mice 7 days after allogeneic BMT showed stronger evidence of lesions such as marked crypt destruction, greater villus blunting, loss of enterocyte brush border, luminal sloughing of cellular debris, and more extensive crypt cell apoptosis in wild-type mice as compared with TLR9^{-/-} mice (Fig. 3, A and B). The severity of lesions was reduced, although not so markedly as in the small intestine, also in the large bowel mucosa of TLR9^{-/-} mice (Fig. 3, C and D), while in the liver and skin the differences were not significant (not shown).

Experiments using BALB/c mice as recipients and C57BL/6 wild-type, TLR9^{-/-}, or TLR4^{-/-} mice as donors revealed no significant differences in survival or GVHD clinical score (see Supplemental Data).⁵

Reduced proliferation of donor-derived T cells after BMT in recipient TLR9^{-/-} mice

Experiments to compare donor cell engraftment in wild-type and TLR9^{-/-} mice after BMT showed that spleen of C57BL/6 wild-type mice contained $73.5 \pm 6.4\%$ (mean $\pm$ SD) Dd+ donor-derived cells and $25.5 \pm 6.3\%$ Db+ recipient cells at day 4 post-transplantation and $92.0 \pm 1.4\%$ and $9.5 \pm 0.7\%$, respectively, at day 7. A slower expansion of donor cells and increased persistence of recipient cells were observed in spleen of TLR9^{-/-} mice, since $21.0 \pm 2.8\%$ of splenocytes were Dd+ and $78.0 \pm 2.9\%$ were still Db+ by day 4. Only at day 14 after BMT did Dd+ cells in knockout mice reach $\sim 90\%$ ($96.5 \pm 0.7\%$ of Dd+ and $2 \pm 1.4\%$ of Db+). Thus, the protection from acute lethal GVHD observed in mice lacking TLR9 is associated with a slower in vivo expansion of alloreactive donor cells and a slower progressive reduction of

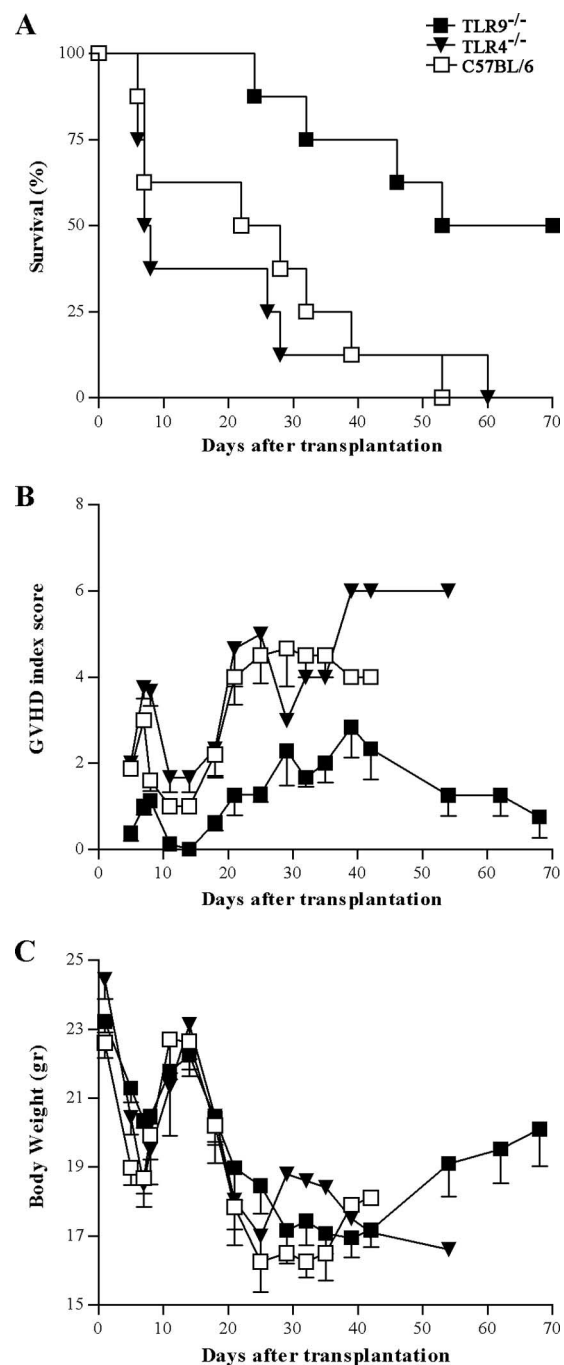


FIGURE 1. Evaluation of acute GVHD in TLR9^{-/-} and TLR4^{-/-} mice. GVHD mortality and morbidity in lethally irradiated (900 cGy) recipient C57BL/6 (□), TLR4^{-/-} (▼), and TLR9^{-/-} (■) mice (eight animals/group) transplanted with 10^7 bone marrow cells and 4×10^7 splenocytes from BALB/c donor mice. Percent survival (A), mean GVHD index score (B), and mean body weight (C) are represented. Differences in survival were calculated by the log-rank test ($p = 0.0062$ TLR9^{-/-} vs C57BL/6; $p = 0.0035$ TLR9^{-/-} vs TLR4^{-/-}); differences in mean index score were evaluated by unpaired t test ($p < 0.0001$ TLR9^{-/-} vs C57BL/6 and TLR4^{-/-}).

lethally irradiated recipient cells. To evaluate the ex vivo proliferation activity of allogeneic donor BALB/c T cells in recipient wild-type and TLR9^{-/-} mice, spleen cell samples were harvested from the two strains 4 days after BMT and enriched for T cells by negative selection with CD19 microbeads. The enriched samples, containing more than 70% CD3⁺ cells, were labeled with CFSE,

⁵ The online version of this article contains supplemental material.

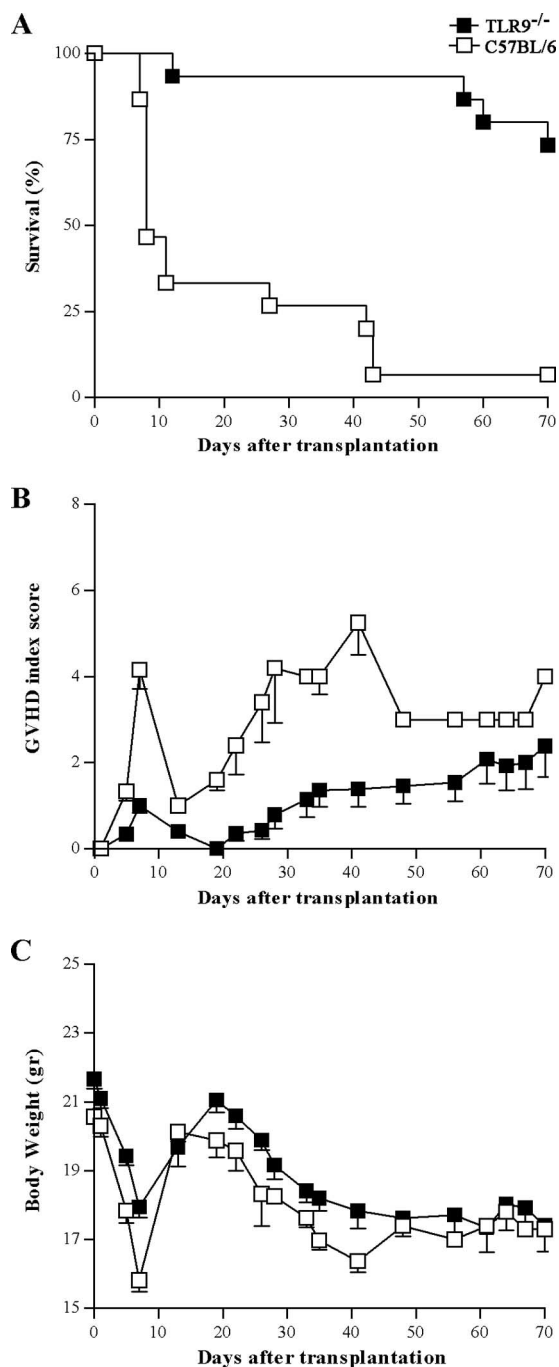


FIGURE 2. Acute GVHD in antibiotic-treated TLR9^{-/-} mice. GVHD mortality and morbidity in lethally irradiated (900 cGy) recipient C57BL/6 (□) and TLR9^{-/-} (■) mice (15 animals/group) that received gentamicin in their drinking water beginning 7 days before and up to 2 mo after transplantation with 10^7 bone marrow cells and 4×10^7 splenocytes from BALB/c donor mice. Data shown were pooled from two separate experiments performed with eight and seven mice per group, respectively. Percent survival (A), mean GVHD index score (B), and mean body weight (C) are represented. Log-rank test: $p < 0.0001$ TLR9^{-/-} vs C57BL/6 for survival; unpaired t test: $p < 0.0001$ TLR9^{-/-} vs C57BL/6 for mean index score.

cultured for 3 days, stained with H2d-specific Ab to identify BALB/c-derived cells, and analyzed by FACScan. As compared with wild-type mice, the donor-derived H2d⁺ population from TLR9^{-/-} transplanted mice contained an increased percentage of nonproliferating CFSE^{high} cells and a reduced number of prolifer-

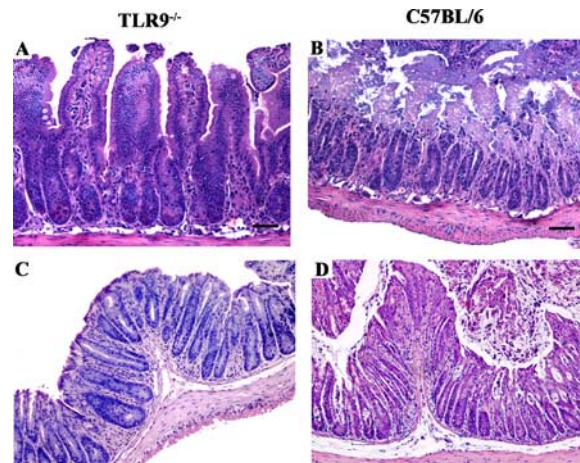


FIGURE 3. Histologic analysis of wild-type and TLR9^{-/-} transplanted mice. Small intestine (A and B) and colon (C and D) tissues from TLR9^{-/-} and C57BL/6 wild-type analyzed 7 days after transplantation with 10^7 bone marrow cells and 4×10^7 splenocytes from BALB/c donor mice. Representative samples from each group of mice were processed and stained with H&E. Scale bars: 60 μ m.

ating CFSE^{low} cells (mean $\pm$ SD CFSE^{high} H2d⁺: $40.6 \pm 3.9\%$ in TLR9^{-/-} vs $21.5 \pm 0.6\%$ in C57BL/6, $p = 0.02$; CFSE^{low} H2d⁺: 27.0 ± 0.5 in TLR9^{-/-} vs 45.6 ± 0.2 in C57BL/6, $p = 0.0004$) (Fig. 4). To monitor in vivo donor cells proliferation, experiments were performed injecting 1.5×10^7 CFSE-labeled BALB/c splenocytes enriched of T cells at day 4 after BMT in C57BL/6 wild-type and TLR9^{-/-} mice. TLR9^{-/-} CFSE-labeled H2d⁺ cells, retrieved from spleens 24 h after injection, showed a reduced percentage of proliferating CFSE^{low} cells as compared with those from C57BL/6 (mean $\pm$ SD proliferating CFSE^{low} cells: $2.2 \pm 1.0\%$ in TLR9^{-/-} vs $12.4 \pm 0.4\%$ in C57BL/6, $p = 0.0063$; non-proliferating CFSE^{high} cells: 95.9 ± 0.5 in TLR9^{-/-} vs 86.4 ± 1.6 in C57BL/6, $p = 0.0161$).

Role of bone marrow- and non-bone marrow-derived TLR9-expressing cells in the pathogenesis of GVHD

CpG-containing bacterial DNA, released from intestinal bacteria upon myeloablative treatment, might enter the circulation through the impaired mucosal barrier and induce activation and maturation of APCs. We therefore evaluated the activation of spleen APCs in C57BL/6 and TLR9^{-/-} mice after myeloablative conditioning. No significant expression of CD80 and CD86 costimulatory molecules on MHC class II⁺ cells was detected in either strain at 3 h after irradiation; these molecules were highly expressed in both strains at 24 h postirradiation, although the up-regulation was less marked in TLR9^{-/-} than in wild-type mice (mean $\pm$ SD CD80: 85.6 ± 1.0 in TLR9^{-/-} mice vs 93.6 ± 0.3 in C57BL/6 mice, $p < 0.0001$; mean $\pm$ SD CD86: 74.5 ± 4.8 in TLR9^{-/-} mice vs 93.0 ± 0.8 in C57BL/6 mice, $p = 0.0006$) (Fig. 5).

In MLR cultures in which T cells from BALB/c mice were stimulated with MHC class II APCs purified from spleens of wild-type or TLR9^{-/-} C57BL/6 mice 24 h after myeloablative conditioning, proliferation of alloreactive BALB/c T cells was significantly lower in cultures stimulated with TLR9^{-/-} MHC class II cells than in cultures stimulated with the wild-type cells (cpm 18919 ± 2008 with TLR9^{-/-} MHC class II cells vs cpm 42131 ± 5728 with C57BL/6 MHC class II cells, mean $\pm$ SD; $p = 0.0082$, at day 3 of culture) (Fig. 6). These differences were not related to a constitutive lower allostimulatory activity of TLR9^{-/-} APCs, since no significant differences in the proliferation of BALB/c T

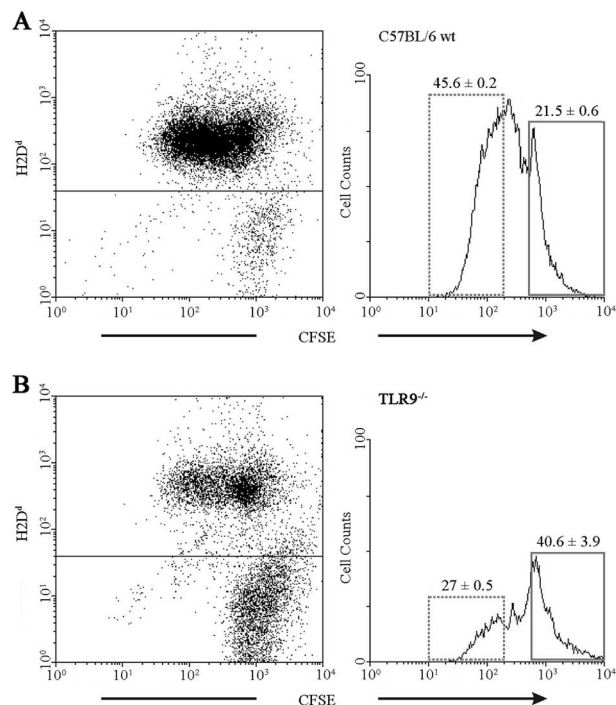


FIGURE 4. Proliferation activity of donor BALB/c cells after BMT in recipient $TLR9^{-/-}$ mice. Proliferation of alloreactive donor cells derived from C57BL/6 wild-type (A) or $TLR9^{-/-}$ (B) mice 4 days after transplantation. Purified T cells were labeled with CFSE, maintained in culture for 3 days, and stained with PE-anti-H2D^d Ab for flow cytometric analysis of the percentage of CFSE^{high}H2D^d and CFSE^{low}H2D^d cells. For analysis, live lymphocytes were gated according to light scatter properties and propidium iodide exclusion. The mean percentage ($\pm$ SD) of CFSE^{high}H2D^d and CFSE^{low}H2D^d cells in one representative experiment is reported; plot and histogram are representative of one mouse per group. Three independent experiments (each with two or three mice per group) with superimposable results were performed.

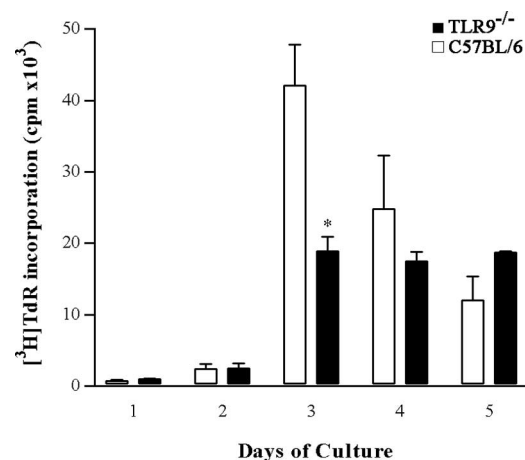


FIGURE 6. Stimulatory activity of $TLR9^{-/-}$ MHC class II cells from myeloablated mice. Proliferation of BALB/c responder T cells cocultured with stimulator MHC class II APCs derived from spleen of wild-type C57BL/6 (open bars) or $TLR9^{-/-}$ (closed bars) mice treated the day before with myeloablative irradiation. Proliferation was determined at different times of coculture by [³H]TdR incorporation for the last 18 h. Data are representative of one experiment performed evaluating proliferation of BALB/c responder cells in the presence of stimulator cells from three different mice per group. Two independent experiments were performed with superimposable results. *, $p = 0.0082$ by unpaired t test.

cells stimulated with APCs from nonconditioned $TLR9^{-/-}$ and C57BL/6 wild-type mice were observed (cpm 12783 ± 5707 and cpm 9995 ± 2177 , respectively; mean $\pm$ SD). Moreover, these differences were not related to a lower capability of $TLR9^{-/-}$ APCs to respond to other TLR or not-TLR maturation stimuli, as revealed by no significant differences in the proliferation of BALB/c T cells stimulated with $TLR9^{-/-}$ and C57BL/6 wild-type APCs pretreated in vitro with LPS, TNF- α , or LPS plus TNF- α .

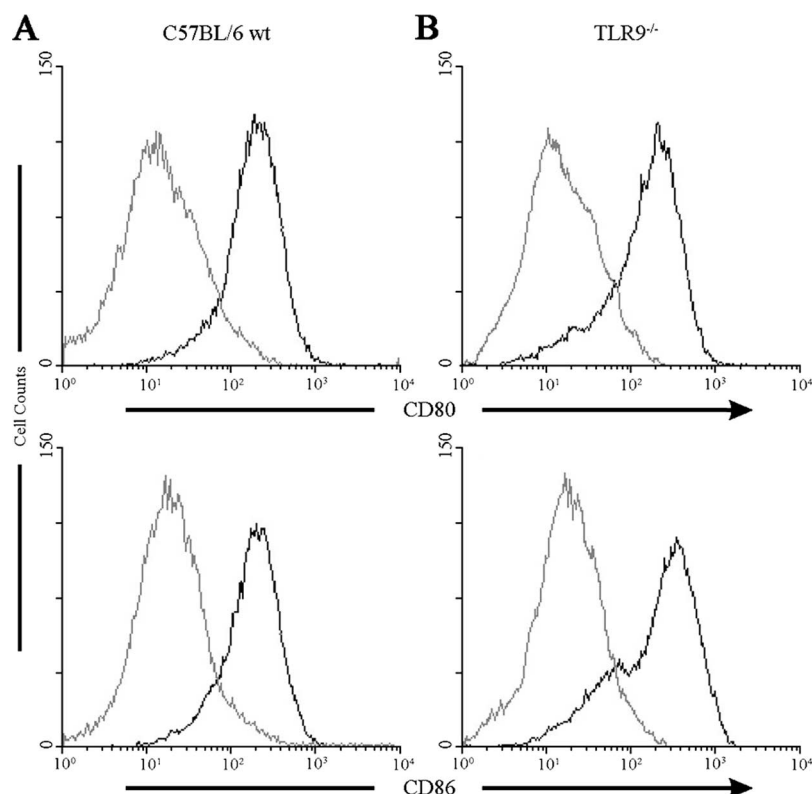


FIGURE 5. Up-regulation of costimulatory markers on MHC class II APCs from C57BL/6 wild-type or $TLR9^{-/-}$ mice after myeloablative irradiation. Expression of CD80 and CD86 in MHC class II APCs obtained from C57BL/6 (A) or $TLR9^{-/-}$ (B) mice 3 (gray line) or 24 h (black line) after myeloablative irradiation. Cells were purified from spleen by MHC class II (Ia) microbeads, stained with FITC-anti-Ia Ab and PE-anti-CD80 or -CD86 Abs, and analyzed by flow cytometry. Analysis was performed on live cells gated on the Ia⁺ population. Plots are representative of one mouse per group. Two independent experiments, with two mice each, with superimposable results were performed.

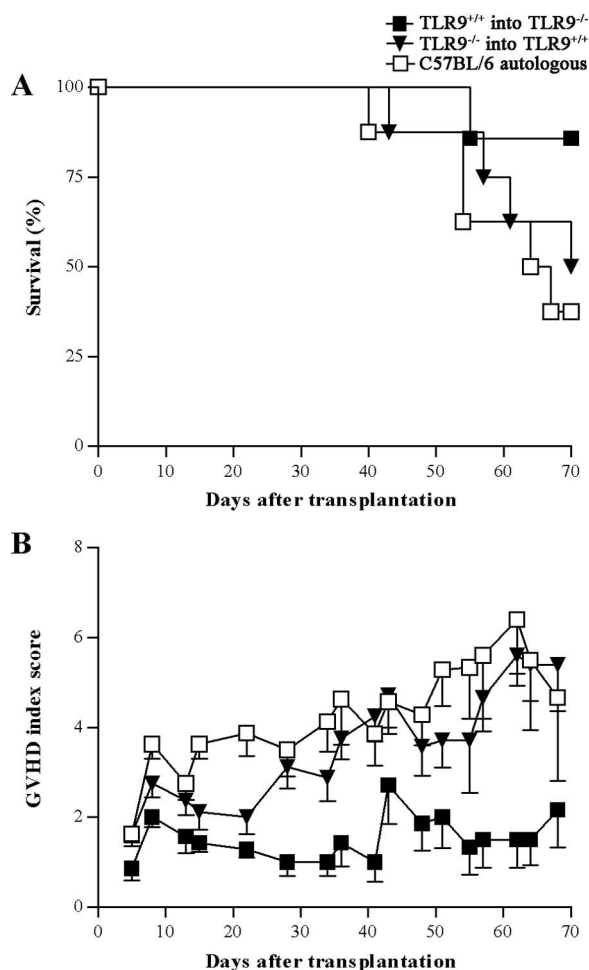


FIGURE 7. Evaluation of acute GVHD in chimeric mice. GVHD mortality and morbidity in lethally irradiated (900 cGy) recipient C57BL/6 autologous transplanted (□) (8 mice), TLR9^{+/+} into TLR9^{-/-} chimeras (■) (7 mice), or TLR9^{-/-} into TLR9^{+/+} chimeras (▼) (8 mice) transplanted with 10^7 bone marrow cells and 4×10^7 splenocytes from BALB/c donor mice. Mice received gentamicin in their drinking water beginning 7 days before and up to 2 mo after transplantation. The percent survival (A) and the mean GVHD index score (B) are represented. Differences in mean index score were evaluated by unpaired *t* test ($p < 0.0001$ TLR9^{+/+} into TLR9^{-/-} chimeras vs C57BL/6 autologous transplanted and TLR9^{-/-} into TLR9^{+/+} chimeras).

(cpm 26804 ± 3006 with LPS-matured-, cpm 21330 ± 6105 with TNF- α -matured-, cpm 27674 ± 8128 with LPS plus TNF- α -matured-TLR9^{-/-} MHC class II cells vs cpm 25348 ± 3803 with LPS-matured-, cpm 16018 ± 4574 with TNF- α -matured-, cpm 24777 ± 1093 with LPS+TNF- α -matured-C57BL/6 MHC class II cells, at day 3 of culture; mean $\pm$ SD).

In addition to activate circulating APCs, bacterial breakdown products of intestinal microflora after conditioning regimens might interact and activate TLR9 expressing intestinal cells, which in turn might release proinflammatory cytokines and chemokines able to recruit and activate allogeneic cells to target intestinal tissue.

To investigate the contribution of non-bone marrow-derived TLR9-expressing cells in the pathogenesis of GVHD, chimeric mice were generated in which: 1) lethally irradiated TLR9^{-/-} mice were reconstituted with TLR9^{+/+} bone marrow cells to express TLR9 only in hematopoietic cells and 2) lethally irradiated TLR9^{+/+} mice were reconstituted with TLR9^{-/-} bone marrow cells to express TLR9 only in nonhematopoietic cells. As a control

group, wild-type mice were autologously transplanted. After checking for complete reconstitution with donor cells, seven TLR9^{+/+} into TLR9^{-/-} chimeras, eight TLR9^{-/-} into TLR9^{+/+} chimeras, and eight autologous-transplanted wild-type mice were myeloablative-irradiated and injected with 10^7 bone marrow cells and 4×10^7 splenocytes from BALB/c donors. As already observed (28), mice that previously underwent to autologous transplant showed a reduced GVHD when subsequently transplanted with allogeneic bone marrow; therefore, a very low dose of donor splenocytes (10^3 cells) was infused three times at 10-day intervals beginning on day 40 from allogeneic transplantation.

In TLR9^{+/+} into TLR9^{-/-} chimeras, the GVHD clinical score was significantly lower than that observed in TLR9^{-/-} into TLR9^{+/+} chimeras or in the control group ($p < 0.0001$) (Fig. 7B). No significant differences were detected in the severity of GVHD between TLR9^{-/-} into TLR9^{+/+} chimeras and the control group. Five of eight control mice and four of eight TLR9^{-/-} into TLR9^{+/+} chimeras succumbed within 70 days, while only one of seven TLR9^{+/+} into TLR9^{-/-} chimeras was dead at 70 days (Fig. 7A).

Discussion

The present study points to a previously undescribed function of TLR9 in the pathogenesis of murine acute GVHD. The improved survival and clinical score of acute GVHD observed in TLR9^{-/-} C57BL/6 recipients of allogeneic BALB/c bone marrow cells were initially associated with reduced engraftment, increased persistence of peripheral blood recipient cells and reduced proliferation activity of BALB/c-derived donor T cells compared with the observations in wild-type mice. After this initial reduction of the engraftment, TLR9^{-/-} mice achieved complete immune reconstitution, with donor-origin PBL reaching 96.5% at day 14 and 100% by day 70.

The significant loss of weight in the presence of a reduced GVHD index score and increased survival observed in TLR9^{-/-} mice could be related, as suggested by Kim et al. (29), to a GVHD confined mainly to the lymphoid compartments. This hypothesis is supported by the histological analyses of target tissues of acute GVHD that revealed reduced gut damage in TLR9^{-/-} mice. Reverse experiments using BALB/c mice as recipients and TLR9^{-/-} mice as donors indicated that expression of TLR9 plays a role only in the recipients. No difference in GVHD morbidity and mortality was observed in recipient mice lacking TLR4, which specifically binds LPS ligand, as compared with wild-type mice, and no increase in survival rate was detected in TLR9/4 double-knockout mice, as compared with TLR9^{-/-} mice, suggesting that in recipients TLR4 does not play a significant role in the pathogenesis of GVHD. Moreover, differently to the reduced acute GVHD in (C3FeB6) F1 recipient mice, using as donor C3H/HeJ mice, which have a point mutation in TLR4 domain, observed by Cooke et al. (30, 31), we did not find that resistance to LPS in C57BL/6 TLR4^{-/-} mice used as donors affected the severity of acute GVHD in the allogeneic BMT.

MHC class II cells isolated from spleen of myeloablated TLR9^{-/-} mice showed a reduced percentage of cells expressing costimulatory molecules and a significantly lower allostimulatory ability than the cells from irradiated wild-type mice, although the differences between stimulatory function of TLR9^{-/-} and wild-type APCs did not appear so high to completely explain the in vivo effect. Experiments performed in chimera mice, showing a significant reduction in GVHD-related morbidity in TLR9^{+/+} into TLR9^{-/-} chimeras, which express TLR9 only in hematopoietic cells, but not in the TLR9^{-/-} into TLR9^{+/+} chimeras, which express TLR9 only in non-bone marrow-derived cells, point to a predominant role of non-bone marrow-derived cells. These cells

might directly participate in the process of Ag presentation (32); accordingly, intestinal cells have been reported to express an array of Ag-presenting molecules, including both classical MHC class I and II molecules and nonclassical class I molecules, which enable intestinal cells to directly present Ags and provide costimulatory signals that influence mucosal immune response (33). Alternatively or in addition, activated intestinal non-bone marrow-derived cells might release proinflammatory cytokines and chemokines able to activate resident APCs (13, 34) and/or to favor recruitment of allogeneic cells to target intestinal tissue, increasing GVHD-related intestinal damage. The induction of tissue inflammation through administration of TLR agonist has been demonstrated to represent a checkpoint of overriding importance in the recruitment of GVHD reactive T cells to peripheral tissue (11) and CpG-oligonucleotides (ODNs) administered at the time of BMT has been reported to accelerate GVHD-induced mortality, increasing donor T cells number in both lymphoid and nonlymphoid parenchymal and epithelial GVHD target tissues (12).

If activation is also mediated by non-bone marrow-derived intestinal cells, the different role of TLR9 and TLR4 in the pathogenesis of acute GVHD might rest in the characteristic features of these receptors on gastrointestinal tract epithelial-origin cells. Intestinal epithelial cells express extremely low levels of TLR4, possibly reflecting a down-regulatory adaptation to the continual stimulation by LPS from commensal bacteria (35, 36), by contrast, as above reported, TLR9 is expressed on intestinal epithelial cells (17–19) and plays a unique role in maintenance of colonic homeostasis, in inflammation, and in regulating tolerance to other TLR ligands (17). Indeed, polymorphisms in TLR9 have been associated with Crohn's disease (37) and TLR9 signaling has been shown essential in mediating the anti-inflammatory effects of probiotics and CpG-ODNs in models of induced experimental colitis (38, 39). On the contrary, a severe exacerbation of inflammatory disease was induced by CpG-ODN when treatment was performed after the establishment of colitis (40) and a reduced inflammatory response to bacterial sepsis has been recently observed in TLR9^{-/-} mice (21). An explanation for this apparent discrepancy in the control of intestinal inflammation by TLR9 is suggested by a recent study which demonstrated the expression of TLR9 on both apical and basolateral surface of intestinal epithelial cells and a different response to apical or basolateral signals: while apical stimulation of TLR9 limited inflammatory response and confers tolerance to other TLR agonists, basolateral stimulation induced inflammation (17).

Thus, histological evidences in TLR9^{-/-} mice as well as the reduced GVHD index score in TLR9^{+/+} into TLR9^{-/-} chimera suggest that gut toxicity might be related to intestinal TLR9 expression and that the gastrointestinal pathology plays a central role in the overall GVHD severity, according to previous observations (13, 31).

Strategies to specifically inhibit intestinal activation of TLR9 by luminal flora have recently been reported (41). These strategies consist in the use of ODNs containing immune-neutralizing motifs, which act as competitors in interaction with TLR9 (42–44). Although in vitro experiments have widely established the potency and specificity of these synthetic ODNs in inhibiting the immune activation elicited by CpG-ODN, fewer studies have been performed in vivo. Preliminary results from experiments to block activation of TLR9 with ODN 2088 inhibitor in C57BL/6 mice transplanted with allogeneic bone marrow cells (3–5 mice/treatment) showed a reduced intestinal damage in samples of small intestine obtained from mice i.p. treated on days 1–3 or on days 3–5 after BMT with ODN 2088 (80 µg); on the contrary, no protection was

observed when treatment with inhibitor was performed before BMT (data not shown).

Although further studies are needed to determine the dose, route, and timing of administration to improve the efficacy of inhibitors, our results indicating major role for TLR9 activation in the pathogenesis of GVHD provide the rationale to the development of such strategies to reduce GVHD.

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Disclosures

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